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Willard Robinson, PhD student, Department of Economics, North Carolina State University

I am an economist who studies the valuation and allocation of public and environmental resources. Mechanisms and valuations always maintain assumptions: zero transaction cost, constant biological stocks, and strategic behavior are three that are relevant to my own research. I study the resilience of those mechanisms and the value of these environmental resources when these assumptions are relaxed. I pursue these questions using economic theory, structural welfare estimation, and reduced-form causal inference. My dissertation studies carbon offset markets and outdoor recreation, but my interests span environmental economics and mechanism design more broadly.

Asymmetric information and costly participation in voluntary conservation markets (Job market paper). Over 24% of the world’s greenhouse gas emissions (World Bank, 2026) are tied to an emissions trading system (such as cap-and-trade). However, substantial emissions abatement opportunities exist in forestry, agriculture, and other nature-based activities, which remain outside these regulated spaces. The offset market is the mechanism by which these outside abatement sources can be incentivized. The common tool in this literature is to assign a baseline level of abatement, and to pay a price for each unit of abatement above this baseline. Nearly every theoretical treatment of these markets (and the ensuing policy prescriptions) assumes frictionless trade. I extend the standard models of such “policy-generated” offset markets to allow for transaction costs, and show that these markets can create negative welfare when the combination of adverse selection and transaction costs is severe enough. This counters three results previously held in the literature: that offset markets are innocuous at worst, that increasing the baseline improves efficiency, and that price does not affect the “additionality” of offsets (whether the offsets represent true abatement). The result requires both asymmetric information and transaction costs. If either of these is corrected, the standard conclusions in the literature apply and offset markets do become truly innocuous.

Until now, the literature has agreed that welfare is increasing in this baseline; the intuition is that requiring agents to provide more abatement upfront leads to a lower average cost for each unit of abatement. This is reversed when transaction costs are present, as higher baselines tend to select agents with higher transaction costs (which creates less—potentially negative—welfare). The literature has also agreed that the price of an offset does not affect the overall “additionality” of the market, and that the optimal price in such a market will never be less than the Pigouvian standard. My paper shows this is not true in the face of transaction costs, and I highlight a case where a sub-Pigouvian price welfare-dominates the market at the Pigouvian price, even when the baseline is tailored for the Pigouvian price.

The largest offset programs in the world combine high, heterogeneous participation costs with unobservable counterfactuals, and the response in the policy sphere has been to try to better measure and ensure additionality. Such advances genuinely improve the regulator’s ability to make policy, but they leave the problem of participation costs—which can be independent of additionality—unaddressed. My paper characterizes when a market is distorted enough that it should be substantially reformed (or even shut down). In doing so, I provide novel results surrounding the baseline’s effect on efficiency, the price’s effect on additionality, and economic value generated by offset markets.

Transaction costs of emissions offsets: Evidence from California (Under review). This paper supplied the original empirical motivation for the theory in my job market paper. High-emitting California firms must surrender permits or offsets against all of their emissions. For the first decade of the program, offsets were priced roughly 30% below permits, yet many firms—in some years, a majority—used no offsets at all. Taken at face value, this appears to be non-cost-minimizing behavior. But the transaction costs required to use offsets rationalize this behavior; previous literature has identified similar costs in the European Union’s emissions trading system. I provide extensive descriptive evidence from raw firm-level compliance data to illustrate this behavior, and I use econometric methods to estimate the value of the transaction cost.

The identification exploits a feature of California’s regulatory policy: the share of compliance that a firm can meet with offsets is capped, and that cap bounds the transaction costs from both sides. The intuition is this: if a firm uses offsets, its transaction cost must be smaller than its cost savings; likewise, if a firm does not use offsets, its transaction cost must be greater than the maximum potential cost savings it could have earned by using them. Building empirical strategies out of institutional details like this cap is representative of the empirical work I look forward to doing in the future.

Outdoor recreation and nonmarket valuation. Nonmarket valuation drew me into economic research, and it remains an active part of my research agenda. My third dissertation chapter, “The impacts of climate-change induced fish migration on angler welfare,” pairs standard nonmarket valuation tools with fish population models drawn from the natural science literature under several climate scenarios. Linking data on shoreline angler trips along the east coast with data on fish stocks and expected climate-dependent future fish stocks lets me trace an indirect channel from sea temperature to fish stocks to catch rates to recreation behavior. This endogenizes fish stocks, a quantity usually held fixed, but one that may substantially shift the value of recreation, particularly in areas further south or north. In related co-authored work, we use a two-decade panel of shoreline anglers (2004–2024) to quantify bimonthly userday values at over 2,600 sites across the coast and apply internal meta-analysis to measure the stability of these values across time and space.

Impacts of centralized budget policy on national parks (PERC graduate fellowship, Summer 2025). In this paper, I examine whether a 2004 policy which gave park managers greater control over user-fee revenue crowded out central federal appropriations. To do this, I digitized three decades of National Park Service budget documents, assembling a panel of more than 10,000 park-year observations spanning 1986–2019. Comparing fee-charging parks with non-fee-charging parks through difference-in-differences, trend-break, and synthetic-control designs, I cannot reject the null hypothesis that no systematic crowding out occurred. This result contributes to a larger discussion about local and federal regimes, suggesting that the policy added genuine funding rather than merely reshuffling dollars. This project is representative of the unique data creation I look forward to continuing in my future research, and the topic relates to my interest in mechanisms designed to allocate public goods (in this case, seemingly successfully).

A matching mechanism in the field (work in progress). In a newer co-authored project, topically distinct from my dissertation, but emblematic of the kind of research I like to do, I study NC State’s centralized engineering major-assignment system, which closely resembles the “Boston mechanism” from

matching theory. A common critique of these mechanisms is that they are not strategyproof: students can potentially manipulate their outcome by strategically misreporting their preferences. My co-author and I surveyed engineering students to elicit their preferences when they first came to NC State, and at the time they submitted their preferences to the major-assignment mechanism. We find that (i) more than 95% of students submitted their true preferences to the mechanism, and (ii) about 65% of students changed their minds on what their top choice major was after the first year. (i) is a somewhat unexpected result, and it motivates the theory we then build: under mild conditions about student preference formation—which are defended using (ii)—the Boston mechanism yields equivalent outcomes to a strategyproof mechanism. This project joins two aspects of research that I consider representative of my future work: collecting original data, and explaining outcomes that appear puzzling at first.

Agenda. What ties these projects together is the distance between how mechanisms are designed to perform and how they perform in practice or when certain assumptions are relaxed. Going forward, I plan to continue working at the nexus of environmental economics and mechanism design by creating original datasets to study real-world policies or conduct nonmarket valuation, and to use economic theory when reality appears different from prediction.